

5.1 HW: Maclaurin Series — Take-home (calculator allowed)

Two problems. Answer on lined paper. Show all working. Total: 36 marks. A GDC may be used; show algebraic working before any numerical evaluation.

4. [Maximum mark: 21]

AA HL 2022 May P1 TZ1 Q12

The function f is defined by $f(x) = e^x \sin x$, where $x \in \mathbb{R}$.

(a) Find the Maclaurin series for $f(x)$ up to and including the x^3 term. [4]

(b) Hence, find an approximate value for $\int_0^1 e^{x^2} \sin(x^2) dx$. [4]

The function g is defined by $g(x) = e^x \cos x$, where $x \in \mathbb{R}$.

(c) (i) Show that $g(x)$ satisfies the equation $g''(x) = 2(g'(x) - g(x))$. [4]

(ii) Hence, deduce that $g^{(4)}(x) = 2(g'''(x) - g''(x))$. [1]

(d) Using the result from part (c), find the Maclaurin series for $g(x)$ up to and including the x^4 term. [5]

(e) Hence, or otherwise, determine the value of

$$\lim_{x \rightarrow 0} \frac{e^x \cos x - 1 - x}{x^3}.$$

[3]

5. [Maximum mark: 15]

AA HL 2025 Nov P3 TZ3 Q02 (a)–(e)

The following problem uses Maclaurin series to investigate approximations of mathematical constants and the accuracies of such approximations.

(a) Given $|x| < 1$, find the sum to infinity of the geometric series

$$1 - x^2 + x^4 - x^6 + \cdots .$$

[2]

(b) Hence, use integration to show that the Maclaurin series of $\arctan x$ may be expressed as

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots .$$

[3]

(c) Using $x = \frac{1}{\sqrt{3}}$ and the first **three** (non-zero) terms of the Maclaurin series of $\arctan x$, find an approximation for π to three decimal places.

[3]

The Maclaurin series of $\arctan x$ is an alternating series.

Theorem. For an alternating series whose terms decrease in magnitude, the error obtained in using a finite number of terms is less than or equal to the absolute value of the next term in the sequence.

By this theorem, the maximum error in using the first three (non-zero) terms as an approximation to $\arctan x$ is $\left| -\frac{x^7}{7} \right|$, that is,

$$\left| \arctan x - \left(x - \frac{x^3}{3} + \frac{x^5}{5} \right) \right| \leq \left| -\frac{x^7}{7} \right|.$$

(d) Determine how many (non-zero) terms of the series would need to be used so that the error in approximating $\arctan\left(\frac{1}{\sqrt{3}}\right)$ is less than 0.0001.

[3]

(e) By using integration by parts, show that

$$\int_0^{1/\sqrt{3}} \arctan x \, dx = \frac{\pi}{6\sqrt{3}} - \frac{1}{2} \ln \frac{4}{3}.$$

[4]